



Department of
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Factor of Safety Electric Jet-Driven Surfboard

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Executive Summary

Project Objective

This analysis determines whether the fiberglass/epoxy composite shell of a jet-driven electric surfboard possesses adequate structural margin against fatigue failure under the combined bending and torsional loads imposed during normal operation. The analysis follows methods from Shigley's Mechanical Engineering Design and applies conservative assumptions throughout to account for the limited published fatigue data on fiberglass/epoxy composite systems.

Loading and Structural Model

The board (length = 2,413 mm) was modeled as a simply supported beam with two concentrated loads: the combined board and rider weight of 844 N (86 kg total) and the propulsion unit weight of 334 N (34 kg). Using beam superposition, the maximum bending moment was calculated to be 477 N·m, occurring at the rider load location, with a maximum deflection of 0.0737 mm — negligible relative to board dimensions. The critical cross-section was modeled as a solid rectangle (609.6 mm wide $\times$ 177.8 mm tall), yielding an area moment of inertia of $2.86 \times 10^{-4} \text{ m}^4$. Shaft torque was derived from the motor operating at 90% of peak output (22.5 kW at 8,100 RPM), producing a steady torque of 26.53 N·m.

Material Properties and Endurance Limit

Material testing established an ultimate tensile strength of $S_{ut} = 91.93 \text{ MPa}$ for the composite. Because fiberglass/epoxy composites lack abundant fatigue research, the corrected endurance limit was taken conservatively as $S'_e = (0.3) S_{ut} = 27.58 \text{ MPa}$, compared to the standard $(0.5) S_{ut}$ used for steels. The modified endurance limit was then computed using Marin correction factors: surface factor $k_a = 0.90$ (machined finish), size factor $k_b = 0.633$ (evaluated at upper bound of 254 mm equivalent diameter), loading factor $k_c = 1.0$, temperature factor $k_d = 1.0$, and reliability factor $k_e = 0.897$ (90% reliability). The final modified endurance limit of $S_e = 14.09 \text{ MPa}$.

Stress Concentration and Conservative Assumptions

Fatigue stress concentration factors were applied to represent the propulsion unit exit cutout in the shell, using mill-keyseat values as a conservative geometric analog: $K_f = 2.14$ (bending) and $K_{fs} = 3.0$ (torsion). Full notch sensitivity ($q = 1$) was assumed throughout due to insufficient composite-specific data, ensuring the analysis does not underestimate the stress concentration effect.

Results

Fatigue analysis was performed using both the DE-Gerber and DE-Goodman distortion energy criteria. The resulting factors of safety were:

- DE-Gerber $\rightarrow n = 22.21$
- DE-Goodman $\rightarrow n = 21.82$

Both values far exceed the minimum acceptable factor of safety of 1.0, confirming the shell has substantial structural margin against fatigue failure under the defined operating conditions.

1 Factor of Safety - Fiberglass/Epoxy Shell

1.1 Assumptions

- The board is modeled as a simply supported beam of length $l = 2413 \text{ mm}$, with concentrated point loads applied at the rider position and propulsion unit location.
- The rider is centered on the board; their weight is divided equally between two support reactions.
- The propulsion unit is mounted at a fixed position along the board's length.
- The fiberglass/epoxy composite is treated conservatively as a homogeneous, isotropic material due to the limited available fatigue research on this material.
- Surface finish is assumed equivalent to a "Machined" finish for the purposes of the surface modification factor, based on published equation parameters.
- A 90% reliability target is selected, consistent with preliminary mechanical design practice for a non-safety-critical prototype.
- The motor operates at 90% of peak output to avoid driving electrical components to their absolute limits.
- Notch sensitivity $q = 1$ is applied conservatively, treating the composite as fully notch-sensitive.
- The endurance limit correction for the composite is taken as $S'_e = (0.3) S_{ut}$ rather than the standard $S'_e = (0.5) S_{ut}$ to account for material uncertainty.
- Loading is treated as fully reversed bending ($M_m = 0, T_a = 0$); steady torque is applied from the driveshaft.

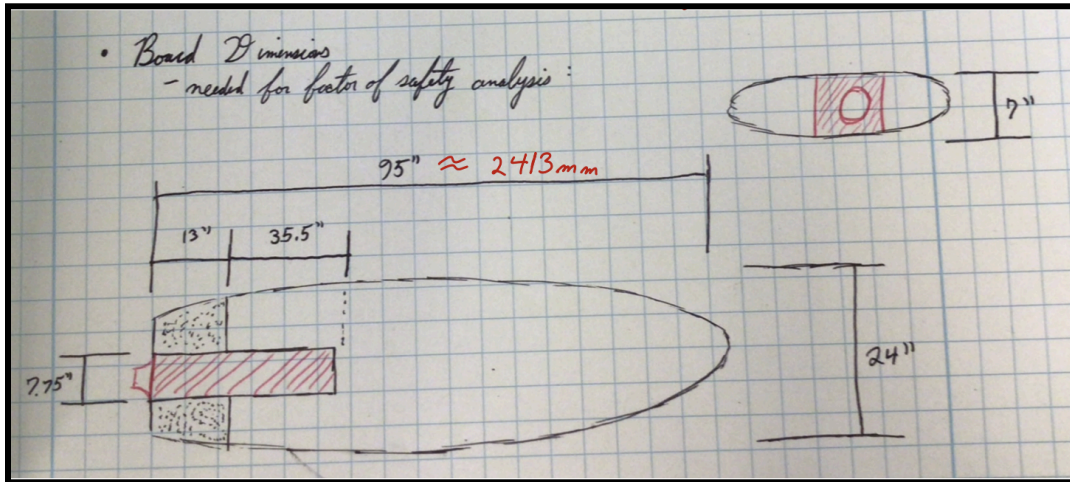


Figure X: Board Dimensions

1.2 Applied Loads

The two primary load sources are (1) the combined weight of the board and rider, and (2) the weight of the propulsion unit.

Board and Rider Weight:

$$W = (m_b + m_r)(g)$$

where:

- m_b = board mass

- m_r = rider mass
- $g = 9.81 \frac{m}{s^2}$

$$W = (m_b + m_r)(g) = (86 \text{ kg})\left(9.81 \frac{m}{s^2}\right) = 843.66 \text{ N} \approx 844 \text{ N}$$

This load is divided equally between two reaction points:

$$F_1 = \frac{W}{2} = \frac{844}{2} = 422 \text{ N}$$

Propulsion Unit Weight:

$$F_p = m_p \cdot g$$

where:

- $m_p = 34 \text{ kg}$ is the mass of the propulsion unit.

$$F_p = m_p \cdot g = (34 \text{ kg})\left(9.81 \frac{m}{s^2}\right) = 333.54 \text{ N} \approx 334 \text{ N}$$

1.3 Section Properties and Bending Moment

Cross-Sectional Moment of Inertia

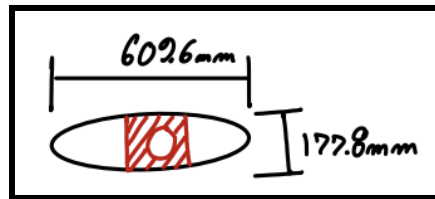


Figure X - Board Cross-Section Dimensioning

The critical cross-section is modeled as a solid rectangle with a width $b = 609.6 \text{ mm}$ and height $h = 177.8 \text{ mm}$. The area moment of inertia about the neutral axis is:

$$I = \frac{1}{12}bh^3$$

$$I = \frac{1}{12} (0.6096 \text{ m})(0.1778 \text{ m})^3 = 2.86 \times 10^{-4} \text{ m}^4$$

Bending Moment Calculation (Superposition)

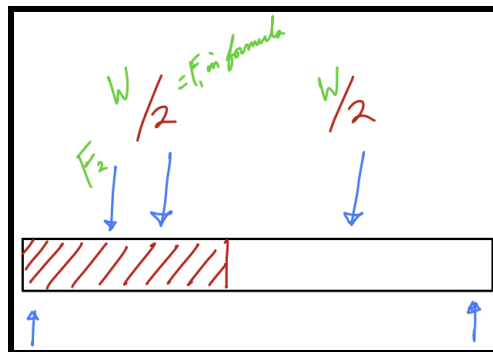


Figure X - Simple Supports Diagram

Using beam superposition (Shigley's Table A-9), the bending moment at each load location is calculated separately and then combined.

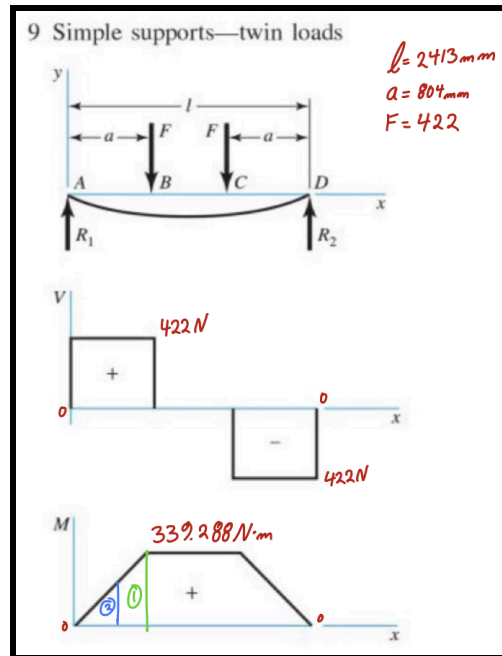


Figure X - Load Case 1

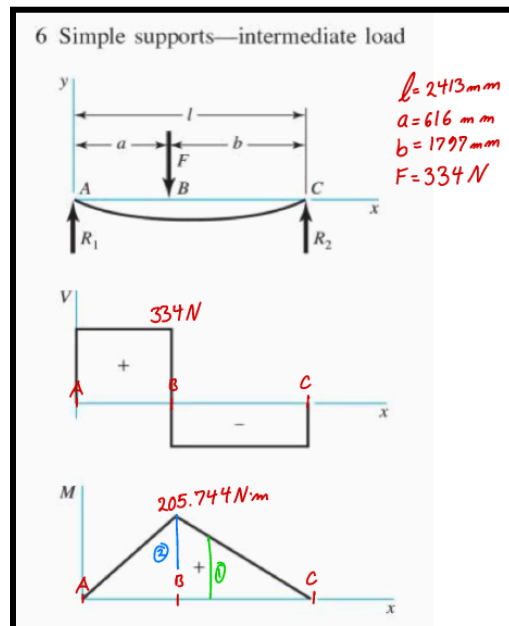


Figure X - Load Case 2

where:

- $a_1 = 804 \text{ mm}$
- $a_2 = 616 \text{ mm}$
- $F_1 = 422 \text{ N}$

- $F_2 = 334 \text{ N}$
- $l = 2413 \text{ mm}$
- $b_2 = 1797 \text{ mm}$

Load Case 1 - Rider / Board Weight

$$M_1 @ 804 \text{ mm} = F_1 a_1 + \frac{F_2(a_2)}{l}(l - a_1) = 476.48 \text{ N} \cdot \text{m}$$

Load Case 2 - Propulsion Unit

$$M_2 @ 616 \text{ mm} = F_1 a_2 + \frac{F_2 \cdot a_2 \cdot b_2}{l} = 413.17 \text{ N} \cdot \text{m}$$

Maximum Bending Moment (Superposition):

$$M_1 = M_{max} \approx 477 \text{ N} \cdot \text{m} = 477,000 \text{ N} \cdot \text{mm}$$

Maximum Deflection

$$y_{max} = 0.0737 \text{ mm}$$

1.4 Shaft Torque

The peak motor output is $P_{peak} = 25 \text{ kW}$. To avoid driving electrical components to their absolute limit, 90% of peak power is used as the operating condition.

Operating Power:

$$P_{90} = (0.90) P_{peak} = (0.90) (25 \text{ kW}) = 22.5 \text{ kW}$$

Operating Shaft Speed:

$$N_{90} = (0.90) (N_{peak}) = (0.90) (9000 \text{ rpm}) = 8100 \text{ rpm}$$

Torque from Power and Speed (Shigley's Eq. 3-43):

$$T = \frac{P_{90} \cdot 60}{2\pi \cdot N_{90}}$$

where:

- T = torque in $\text{N} \cdot \text{m}$
- P_{90} is in W
- N_{90} is in rpm

$$T = \frac{(22,500)(60)}{2\pi(8100)} = 26.53 \text{ N} \cdot \text{m} = 26,530 \text{ N} \cdot \text{mm}$$

1.5 Marin Endurance Modification Factors

The modified endurance limit S_e is computed by applying a series of Marin modification factors to the correct material endurance limit S'_e :

$$S_e = k_a k_b k_c k_d k_e S'_e$$

Surface Factor (k_a)

Surface Finish	Factor a		Exponent b
	S_{ut} , kpsi	S_{ut} , MPa	
Ground	1.21	1.38	-0.067
Machined or cold-drawn	2.00	3.04	-0.217
Hot-rolled	11.0	38.6	-0.650
As-forged	12.7	54.9	-0.758

Figure X - Table 6-2, Curve Fit Parameters for Surface Factor

The Marin surface factor accounts for the difference between the test specimen surface finish and the actual component surface. It is given by (Shigley's Table 6-2):

$$k_a = a \cdot S_{ut}^b$$

where **a** and **b** are empirical constants for the assumed surface finish, and S_{ut} is the ultimate tensile strength of the material.

- Surface finish assumed: Machined (based on Marin curve-fit parameters)
- $a = 3.04 \text{ MPa}$
- $b = -0.217$
- $S_{ut} = 91.93 \text{ MPa}$ (determined from material testing)

$$k_a = (3.04) (91.93)^{-0.217} = 1.13$$

Since k_a cannot physically exceed 1.0, and to remain conservative, the following value is adopted:

$$k_a = 0.9$$

Size Factor (k_b)

$k_b =$	$(d/0.3)^{-0.107} = 0.879d^{-0.107}$	$0.3 \leq d \leq 2 \text{ in}$
	$0.91d^{-0.157}$	$2 < d \leq 10 \text{ in}$
	$(d/7.62)^{-0.107} = 1.24d^{-0.107}$	$7.62 \leq d \leq 51 \text{ mm}$
	$1.51d^{-0.157}$	$51 < d \leq 254 \text{ mm}$

Figure X - k_b Surface Factor

For non-circular cross-sections loading in bending, an equivalent diameter d_e is used (Shigley's Table 6-3):

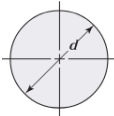
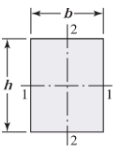
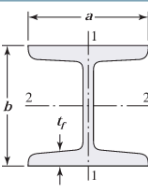
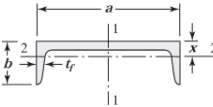
	$A_{0.95} = 0.01046d^2$ $d_e = 0.370d$
	$A_{0.95} = 0.05hb$ $d_e = 0.808 \sqrt{hb}$
	$A_{0.95} = \begin{cases} 0.10at_f & \text{axis 1-1} \\ 0.05ba & t_f > 0.025a \\ & \text{axis 2-2} \end{cases}$
	$A_{0.95} = \begin{cases} 0.05ab & \text{axis 1-1} \\ 0.052xa + 0.1t_f(b-x) & \text{axis 2-2} \end{cases}$

Figure X - Table 6-3, Areas of Common Nonrotating Structural Shapes Loaded in Bending

$$d_e = 0.808 \sqrt{hb}$$

where:

- $b = 609.6 \text{ mm}$
- $h = 177.8 \text{ mm}$

$$d_e = 0.808 \sqrt{(609.6)(177.8)} = 266 \text{ mm}$$

Since $d_e = 266 \text{ mm}$ falls outside the recommended range of the size factor equation (upper bound of 254 mm), k_b is evaluated at the upper bound of $d = 254 \text{ mm}$.

$$k_b = 1.51(d)^{-0.157}$$

$$k_b = 1.51(254)^{-0.157} = 0.633$$

Loading Factor (k_c)

$k_c =$	1	bending
	0.85	axial
	0.59	torsion

Figure X - Loading Factor k_c

$$k_c = 1.0$$

Temperature Factor (k_d)

There is limited quantitative data on the fatigue behavior of fiberglass/epoxy composites under moderate temperature variation within the typical operational range of a watercraft. Therefore, no temperature correction is applied:

$$k_d = 1.0$$

Reliability Factor (k_e)

A 90% reliability level is selected. A 50% reliability target would carry a 50% probability of failure at the calculated stress level, which is unacceptable even for an initial prototype. 99% would be required if failure consequences were catastrophic; 90% is consistent with preliminary mechanical design practice for a non-safety-critical component (Shigley's Table 6-4):

Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702

Figure X - Table 6-4, Reliability Factors k_e
Corresponding to 8 Percent Standard Deviation of the Endurance Limit

$$k_e = 0.897$$

Modified Endurance Limit (S_e)

$$S_e = \begin{cases} 0.5 S_{ut} & S_{ut} \leq 200 \text{ kpsi (1400 MPa)} \\ 100 \text{ kpsi} & S_{ut} > 200 \text{ kpsi} \\ 700 \text{ MPa} & S_{ut} > 1400 \text{ MPa} \end{cases}$$

Figure X - Estimating Endurance Limit

Because fiberglass/epoxy does not have abundant fatigue research, the corrected endurance limit S'_e is taken conservatively as:

$$S'_e = (0.3) S_{ut} = (0.3) (91.93 \text{ MPa}) = 27.579 \text{ MPa}$$

This is more conservative than the standard steel approximation of $(0.5) S_{ut}$, reflecting greater uncertainty in composite fatigue data.

Applying all Marin factors:

$$S_e = k_a k_b k_c k_d k_e S'_e$$

$$S_e = (0.9) (0.633) (1.0) (1.0) (0.897) (27.579)$$

$$S_e = 14.09 \text{ MPa}$$

Stress Concentration Factors (K_f) and (K_{fs})

Stress concentration factors are included to represent the propulsion unit exit cutout in the shell. Values for an mill-keyseat are used as a conservative geometric representation (Shigley's Table 7-1):

	Bending	Torsional	Axial
Shoulder fillet—sharp ($r/d = 0.02$)	2.7	2.2	3.0
Shoulder fillet—well rounded ($r/d = 0.1$)	1.7	1.5	1.9
End-mill keyseat ($r/d = 0.02$)	2.14	3.0	—
Sled runner keyseat	1.7	—	—
Retaining ring groove	5.0	3.0	5.0

Figure X - Table 7-1, First Iteration Estimates for Stress Concentration Factors K_f and K_{fs}

<u>Loading Type</u>	<u>K_t (Theoretical)</u>
Bending	2.14
Torsional	3.0

Figure X - First Iteration K_f and K_{fs} Utilized

The fatigue stress concentration factors K_f and K_{fs} are computed from the notch sensitivity q :

$$K_f = 1 + q (K_t - 1)$$

$$K_{fs} = 1 + q_{shear} (K_{ts} - 1)$$

Due to insufficient research on fiberglass and fiberglass/epoxy composites, notch sensitivity is set to $q = q_{shear} = 1$ (fully notch-sensitive) to treat the analysis conservatively:

$$K_f = 1 + (1) (2.14 - 1) = 2.14$$

$$K_{fs} = 1 + (1) (3.0 - 1) = 3.0$$

1.6 Factor of Safety: DE-Gerber

The Distortion Energy approach uses two amplitude groupings, with $T_a = 0$ and $M_m = 0$:

$$A = \sqrt{4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2}$$

$$A = \sqrt{4 ((2.14) (477))^2 + 0} = 2041.56 \text{ N} \cdot \text{m}$$

$$B = \sqrt{4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2}$$

$$A = \sqrt{0 + 3 ((3.0) (26.53))^2} = 137.854 \text{ N} \cdot \text{m}$$

$$\frac{1}{n} = \frac{8A}{\pi d^3 S_e} \left\{ 1 + \left[1 + \left(\frac{2B S_e}{A S_{ut}} \right)^2 \right]^{1/2} \right\}$$

$$n_{Gerber} = 22.21$$

1.7 Factor of Safety: DE-Goodman

The Distortion Energy approach uses two amplitude groupings, with $T_a = 0$ and $M_m = 0$:

$$\frac{1}{n} = \frac{16}{\pi d^3} \left\{ \frac{1}{S_e} \left[4 (K_f M_a)^2 + 3 (K_{fs} T_a)^2 \right]^{1/2} + \frac{1}{S_{ut}} \left[4 (K_f M_m)^2 + 3 (K_{fs} T_m)^2 \right]^{1/2} \right\}$$

$$n_{Goodman} = 21.82$$

1.8 Factor of Safety Conclusion

Both criteria yield factors of safety well above 1.0, confirming that the fiberglass shell structure has substantial margin against fatigue failure under the defined loading conditions. The conservative material assumptions (reduced S'_e , full notch sensitivity, machined surface penalty) indicate the true margin may be even larger once experimental fatigue data for this specific composite system becomes available.

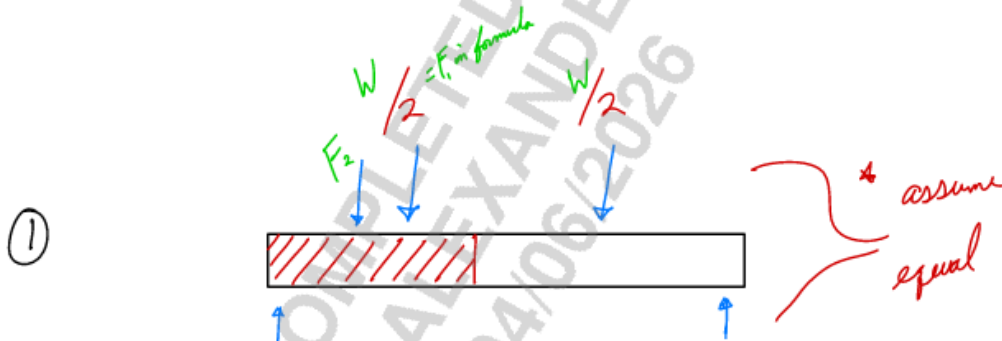
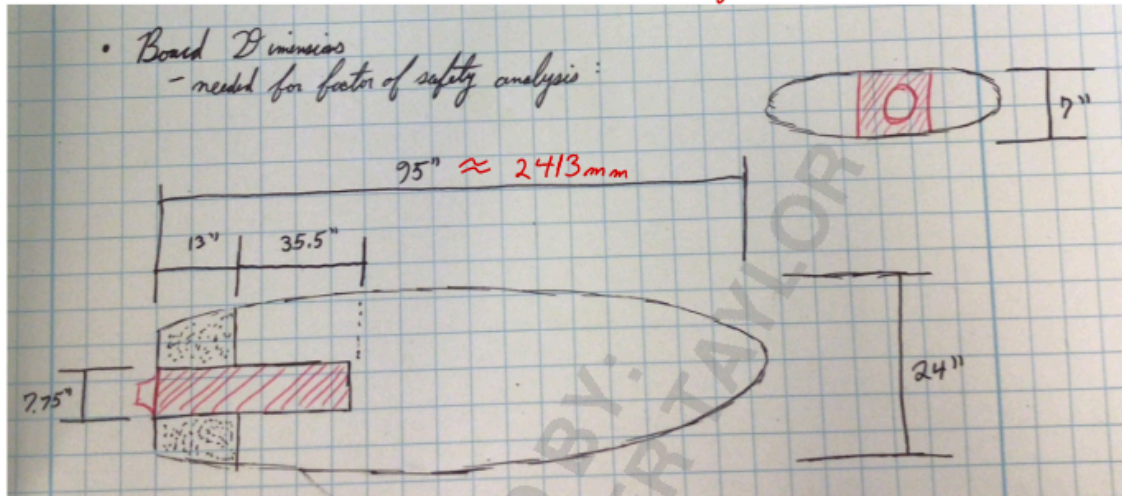
The material testing completed during this project included four-point bending and tensile testing. It established a functional baseline for the fiberglass-epoxy used in the shell, but the scope of testing was constrained by available time and budget. Future work would benefit significantly from a more comprehensive empirical dataset, particularly for fatigue behavior. Currently, the endurance limit used in the factor of safety analysis was estimated conservatively at 30% of the measured ultimate tensile strength $S'_e = (0.3) S_{ut}$ due to the lack of published S-N curve data specific to this composite. A fatigue testing dataset would allow that conservative assumption to be replaced with experimentally validated values, likely revealing a higher usable endurance limit and enabling more efficient material utilization in future designs.

Creating a matrix to include multiple layups, ply counts, fiber orientations, and fiber volume would provide a structured design optimization. Testing specimens fabricated under varying process conditions would quantify the sensitivity of mechanical properties and manufacturing variability.

Impact and shear strength testing would address two failure modes that current testing did not directly characterize but are relevant to a board that experiences hydrodynamic impacts, foot loading, and point loads around mounting hardware. Taken together, a more complete test program would reduce the conservative methods currently embedded in the structural analysis and provide the empirical foundation needed to confidently specify thinner, lighter, or higher-performance layups in subsequent design iterations.

Appendix X - Fiberglass Shell Factor of Safety
Detailed Hand Calculation Work

Factor of Safety - Fiberglass Shell



$$\begin{aligned} \underline{W} &= \text{total weight of board + rider} = (m_b + m_r)(g) \\ &= (9 \text{ kg} + 77 \text{ kg})(9.81 \text{ m/s}^2) \\ &= 843.66 \text{ N} \approx 844 \text{ N} \end{aligned}$$

$$\underline{F_1} = \frac{W}{2} = 422 \text{ N}$$

$$\begin{aligned} \underline{F_2} &= \text{total weight of propulsion unit} = (m_p)(g) \\ &= (34 \text{ kg})(9.81 \text{ m/s}^2) \\ &= 333.54 \text{ N} \approx 334 \text{ N} \end{aligned}$$

② Table A-9 #9 to get M_{max} @ middle and y_{max}

9 Simple supports—twin loads

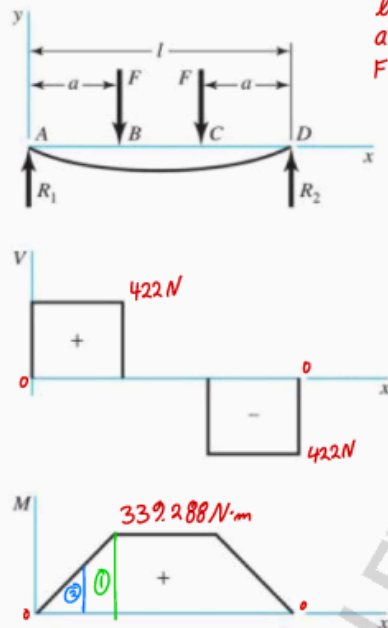
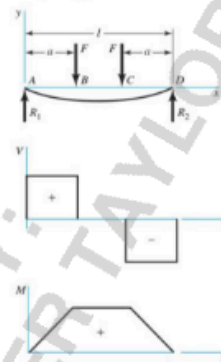


Table A-9 Shear, Moment, and Deflection of Beams (Continued)

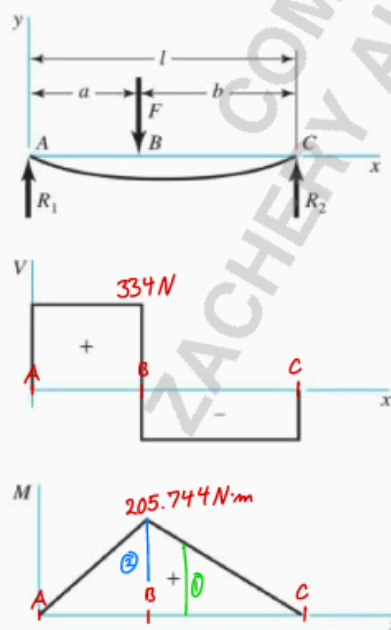
(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 3-2.)

9 Simple supports—twin loads

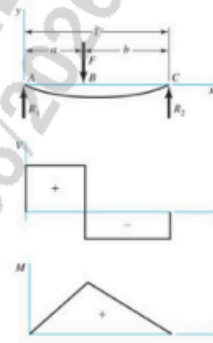


$$\begin{aligned}
 R_1 &= R_2 = F & V_{AB} &= F & V_{BC} &= 0 \\
 V_{CD} &= -F \\
 M_{AB} &= Fx & M_{BC} &= Fa & M_{CD} &= F(l-x) \\
 y_{AB} &= \frac{Fx}{6EI}(x^2 + 3a^2 - 3la) \\
 y_{BC} &= \frac{Fa}{6EI}(3x^2 + a^2 - 3lx) \\
 y_{max} &= \frac{Fa}{24EI}(4a^2 - 3l^2)
 \end{aligned}$$

6 Simple supports—intermediate load



6 Simple supports—intermediate load

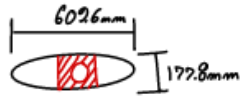


$$\begin{aligned}
 R_1 &= \frac{Fb}{l} & R_2 &= \frac{Fa}{l} \\
 V_{AB} &= R_1 & V_{BC} &= -R_2 \\
 M_{AB} &= \frac{Fbx}{l} & M_{BC} &= \frac{Fa}{l}(l-x) \\
 y_{AB} &= \frac{Fbx}{6EI}(x^2 + b^2 - l^2) \\
 y_{BC} &= \frac{Fa(l-x)}{6EI}(x^2 + a^2 - 2lx)
 \end{aligned}$$

$$a_1 = 804 \text{ mm} \quad F_1 = 422 \text{ N}$$

$$a_2 = 616 \text{ mm} \quad F_2 = 334 \text{ N} \quad b_2 = 1797 \text{ mm}$$

$$l = 2713 \text{ mm} \quad E = 11.8 \text{ GPa}$$



$$I = \frac{1}{12} b h^3 = 0.000286 \text{ m}^4$$

$$M_1 @ 804 \text{ mm} = F_1(a_1) + \frac{F_2(a_2)}{l} (l - a_1)$$

$$M_1 = 476.4791 \text{ N}\cdot\text{m}$$

$$M_2 @ 616 \text{ mm} = F_1(a_2) + \frac{F_2 b(a_2)}{l}$$

$$M_2 = 413.1729 \text{ N}\cdot\text{m}$$

$$M_{\max} \approx 477 \text{ N}\cdot\text{m} \approx 477000 \text{ N}\cdot\text{mm}$$

$$y_{\max} = \frac{F_1(a_1)}{6EI} [(a_1)^2 + 3(a_1)^2 - 3l(a_1)] + \frac{F_2 a_2 (l - a_1)}{6EI l} [(a_1)^2 + (a_2)^2 - 2l(a_1)]$$

$$y_{\max} = 0.0737 \text{ mm}$$

③

T

Peak motor output to driveshaft = $25 \text{ kW} \approx 33 \text{ hp}$

* Will use 90% of peak as "top" to account for capping electrical components

Using formula (3-43)

$$H = \frac{T n}{63025}$$

H = power, hp

n = shaft speed, rev/min

T = torque, lbf·in

* convert to metric *

$$P_{\text{kW}} = \frac{T_{\text{N}\cdot\text{m}} n}{9550} \Rightarrow T_{\text{N}\cdot\text{m}} = \frac{(9550)(P_{\text{kW}})}{n}$$

$$P_{\text{kW},90} = \text{power, kW} = P_{\text{kW}}(90\%) \quad n_{90} = \text{shaft speed, rev/min} = n(90\%)$$

$$T_{90} = \text{torque, N}\cdot\text{m}$$

$$n_{90} = 9000 \text{ rpm}(90\%) = 8100 \text{ rpm} \quad P_{\text{kW},90} = 25 \text{ kW}(90\%) = 22.5 \text{ kW}$$

$$T_{90} = \frac{(9550)(22.5)}{8100} = 26.53 \text{ N}\cdot\text{m}$$

$$\approx 26530 \text{ N}\cdot\text{mm}$$

④ k_a = Surface Factor

Using Table 6-2

$$k_a = a A_{UT}^b$$

A_{UT} = ultimate tensile strength
of the material in MPa

* Surface Finish for this project will be assumed to be
"Machined" based on the Curve Fit Parameters for Surface Factor

$$a = 3.04 \text{ MPa}$$

$$b = -0.217 \text{ [exponent]}$$

$$A_{UT} = \text{determined based on testing data} = 91.93 \text{ MPa}$$

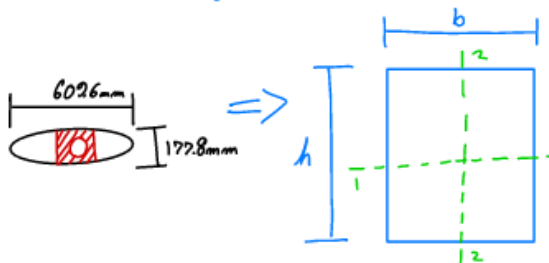
$$k_a = (3.04 \text{ MPa}) (91.93 \text{ MPa})^{-0.217} \approx 1.13$$

k_a cannot be greater than 1.0
for research purposes we will use 0.9 to
be conservative.

$$k_a = 0.9$$

⑤ k_b = Size Factor

Using Table 6-3



$d = d_c$ for Common Nonrotating Structural Shapes Loading in Bending

$$d_c = 0.808 \sqrt{hb}$$
$$= 0.808 \sqrt{(602.6 \text{ mm})(177.8 \text{ mm})}$$
$$d_c = 266 \text{ mm}$$

* Noting variable approximation

Since $d_c = 266 \text{ mm}$ is outside the recommended range, k_b is taken as the value at 254 mm and not reduced further.

Using (6-19)

$$k_b = \begin{cases} (d/7.62)^{-0.107} & 7.62 \leq d \leq 51 \text{ mm} \\ 1.51 d^{-0.157} & 51 \leq d \leq 254 \text{ mm} \end{cases} = 1.24 d^{-0.107}$$

$$k_b = 1.51 (254)^{-0.157}$$

$$k_b = 0.633$$

⑥ k_c = Loading Factor
(6-25)

$$k_c = \begin{cases} 1 & \text{bending} \\ 0.85 & \text{axial} \\ 0.59 & \text{torsion} \end{cases}$$

$$k_c = 1$$

⑦ k_d = Temperature Factor

* Note

There is little quantitative data for how fiberglass/epoxy reacts to moderate temperature fluctuations within the typical working range.

Therefore 1 will be utilized for k_d .

$$k_d = 1$$

⑧

k_e = Reliability Factor

Reliability, %	Transformation Variate z_o	Reliability Factor k_o
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702
99.999	4.265	0.659
99.9999	4.753	0.620

Using 90% Reliability:

50% is too optimistic, with a 50% chance of failure at that stress level — this is not acceptable even for an initial prototype
 >90% would be sufficient where failure has catastrophic consequences.

90% is consistent with preliminary mechanical design practice for a non-safety critical component.

$$\textcircled{9} \quad A_e = k_a k_b k_c k_d k_e (A_e') = k_a k_b k_c k_d k_e \left(\frac{A_{ut}}{2} \right)$$

$$k_a = 0.9$$

$$k_b = 0.633$$

$$k_c = 1$$

$$k_d = 1$$

$$k_e = 0.897$$

$$A_e' = \begin{cases} 0.5 A_{ut}, & A_{ut} \leq 1400 \text{ MPa} \\ 700 \text{ MPa}, & A_{ut} > 1400 \text{ MPa} \end{cases}$$

* Note

To be extensively conservative with a material that does not have abundant research completed on it we will

$$\text{use } A_e' = 0.3 A_{ut} = 27.579 \text{ MPa}$$

$$A_e = (0.9)(0.633)(1)(1)(0.897)(27.579)$$

$$A_e = 14.09$$

⑩ K_f & K_{fs} $\Rightarrow$ First Iteration Estimates for
Stress-Concentration Factors K_t & K_{ts}

* Utilized to represent rear propulsion exit cut

Table 7-1 Bending Torsional

Using End-mill keyseat 2.14 3.0

$$K_f = 1 + g(K_t - 1) \quad K_{fs} = 1 + g_{shear}(K_{ts} - 1)$$

* $g = g_{shear} = 1$;

There is not sufficient research on fiberglass/epoxy. For analysis applying $g=1$ will remain conservative in the analysis; treating the composite material as fully notch-sensitive (no stress relief below K_t)

$$K_f = 1 + g(K_t - 1)$$

$$K_f = 1 + 1(2.14 - 1)$$

$$K_f = 2.14$$

$$K_{fs} = 1 + g_{shear}(K_{ts} - 1)$$

$$K_{fs} = 1 + 1(3.0 - 1)$$

$$K_{fs} = 3.0$$

DE + Gerber

* Use Gerber since first design

Gerber $\frac{n\sigma_a}{S_e} + \left(\frac{n\sigma_m}{S_{ut}}\right)^2 = 1$	
• DE + Gerber	
$\frac{1}{n} = \frac{8A}{\pi d^3 S_e} \left\{ 1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}} \right)^2 \right]^{1/2} \right\}$	$A = \sqrt{4(K_f M_a)^2 + 3(K_{fs} T_a)^2}$
$d = \left(\frac{8nA}{\pi S_e} \left\{ 1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}} \right)^2 \right]^{1/2} \right\} \right)^{1/3}$	$B = \sqrt{4(K_f M_m)^2 + 3(K_{fs} T_m)^2}$

$$A = \sqrt{4[(2.14)(477 \text{ N}\cdot\text{m})]^2} \quad * T_a = 0$$

$$A = 2041.56 \text{ N}\cdot\text{m} \approx 2041560 \text{ N}\cdot\text{mm}$$

$$B = \sqrt{3[(3.0)(26.53 \text{ N}\cdot\text{m})]^2} \quad * M_m = 0$$

$$B = 137.854 \text{ N}\cdot\text{m} \approx 137854 \text{ N}\cdot\text{mm}$$

$$\frac{8A}{\pi d^3 S_e} \approx 0.02251 \quad \left\{ 1 + \left[1 + \left(\frac{2BS_e}{AS_{ut}} \right)^2 \right]^{1/2} \right\} \approx 2.0$$

$$1/n \approx 0.045$$

$$n = 22.21$$

DE + Goodman

- DE + Modified Goodman

$$\frac{1}{n} = \frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}}$$

$$\frac{1}{n} = \frac{16}{\pi d^3} \left\{ \frac{1}{S_e} [4(K_f M_a)^2 + 3(K_{fs} T_a)^2]^{1/2} + \frac{1}{S_{ut}} [4(K_f M_m)^2 + 3(K_{fs} T_m)^2]^{1/2} \right\}$$

$$d = \left(\frac{16n}{\pi} \left\{ \frac{1}{S_e} [4(K_f M_a)^2 + 3(K_{fs} T_a)^2]^{1/2} + \frac{1}{S_{ut}} [4(K_f M_m)^2 + 3(K_{fs} T_m)^2]^{1/2} \right\} \right)^{1/3}$$

* $T_a = 0$; $M_m = 0$

$$\frac{1}{n} = \frac{16}{\pi d^3} \left\{ \frac{1}{S_e} [4(K_f M_a)^2]^{1/2} + \frac{1}{S_{ut}} [3(K_{fs} T_m)^2]^{1/2} \right\}$$

$$\frac{1}{n} = 0.0458$$

$$n = 21.82$$